Want to outsmart your partner's spending spree?

In today's world, financial stability is becoming very important. It is important to understand spending behavior and what affects it. Our study began after a common theme emerged among our friends - their bank balances seemed to be dropping at an alarming rate after spending time with their significant other. This led us to ask the question - How can you outsmart your partner's spending spree? To answer this question, we looked at whether seasonality or type of spending affects expenditure to ensure that our friends can keep their partners happy while keeping their money in the bank.

For our study, we got our data from the website Kaggle.com. The data we downloaded was titled 'Credit Card Spend Analysis'. The file contains variables such as spend amount, expenditure type, month, and year, of 26051 individuals. With a usability score of 9.41out of 10 and a high sample size, our group decided that this would be the perfect dataset to use for our analysis. To prepare the data for analysis, we cleaned the data, which included converting the amount spent from Indian Rupees to US dollars to ensure that our audience would better understand the values.

We first began by taking a look at spending through a wider lens. We believed that men might spend less during winter months due to the practicality of staying indoors. Whereas during the summer they may spend more, paying for outdoor dates and activities. In regards to women, we believed that they might spend more in the summer due to social pressures and the desire to maintain a certain appearance. While in winter, women might spend less, focusing on more practical, cozy clothing and activities that require less preparation and expense. Due to the weather, they may not go out as much leading to a reduction in spending on clothes and accessories. We conducted hypothesis testing to investigate the impact of seasonality on both male and female spending. To begin our analysis, we created dummy variables for seasonality to perform regression effectively. We then defined the null and alternative hypotheses for each gender.

$$H_0: \mu_{spring} = \mu_{summer} = \mu_{autumn} = \mu_{winter}$$

H_a: At least 2 seasons have different in female spending

After our hypotheses were defined, we set our α=0.05 and performed ANOVA Analysis.

SUMMARY

Groups	Count	Sum	Average	Variance
Winter	4104	658915670	160554.5	1.2674E+10
Spring	4044	663150602	163983.828	1.3617E+10
Summer	2025	321309723	158671.468	1.3426E+10
Autumn	3507	561935035	160232.402	1.4168E+10

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	4.9279E+10	3	1.6426E+10	1.22154539	0.3000723	2.60555841
Within Groups	1.839E+14	13676	1.3447E+10			
Total	1.8395E+14	13679				

The ANOVA analysis provided us with the F value of 1.22 and the P value of 0.3. As the P value (0.03) > α=0.05, we were unable to reject the null hypothesis, concluding that there is no significant difference between seasons for females. To further our studies, we conducted regression analysis on the different seasons and spending on SPSS. For this, we created dummy variables for each season and performed regression analysis to find a linear relationship. Our findings showed that the R-squared of the model was only 0.06. This value is extremely small, showing that seasonality can only explain 0.6% of the spending amount, leading us to the conclusion that seasonality does not affect the amount spent by females.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics: R Square Change	F Change	df1	df2	Sig. F Change	Durbin-Watson
1	.080[a]	.006	.006	163495.913	.006	8.539	3	3996	<.001	.866

a. Predictors: (Constant), season=winter, season=summer, season=spring

b. Dependent Variable: spend

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.847E+11	3	2.282E+11	8.539	<.001[b]
	Residual	1.068E+14	3996	2.673E+10		
	Total	1.075E+14	3999			

a. Dependent Variable: spend

b. Predictors: (Constant), season=winter, season=summer, season=spring

We repeated the process to find if seasonality affected male spending. We first filtered the data to only include male spending and then defined our hypothesis :

$$H_0: \mu_{spring} = \mu_{summer} = \mu_{autumn} = \mu_{winter}$$

H_a: At least 2 seasons have different in male spending

After our hypotheses were defined, we set α=0.05 and performed ANOVA Analysis. We made sure to keep the same level of significance to ensure a fair test between men and women. The ANOVA single Factor provided us with the following results :

SUMMARY

Groups	Count	Sum	Average	Variance
Winter	3753	572741925	152609.093	7448365209
Spring	3601	538754187	149612.382	7410106978
Summer	1917	286445858	149424.026	7379824544
Autumn	3101	471580373	152073.645	7519635553

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	2.4839E+10	3	8279760323	1.11220206	0.34262346	2.08423865
Within Groups	9.2073E+13	12368	7444474889			
Total	9.2098E+13	12371				

Our findings show that the F value is 1.11 and the P-value is 0.34. Since P-value =0.34 > α=0.05, we could not reject the null hypothesis showing that there is no significant difference between seasons and expenditure for me. Subsequently, we created dummy variables again for each season to perform regression analysis to conclude whether a linear regression model can be built on SPSS.

Model Summary[b]

					Change Statistics					
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	df1	df2	Sig. F Change	Durbin-Watson
1	.027[a]	.001	.000	85667.087	.001	.985	3	3996	.399	2.030

a. Predictors: (Constant), season=winter, season=summer, season=spring

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2.168E+10	3	7225623757	.985	.399[b]
	Residual	2.933E+13	3996	7338849835		
	Total	2.935E+13	3999			

a. Dependent Variable: Spend
b. Predictors: (Constant), season=winter, season=summer, season=spring

The analysis gave us an R square of 0.001 This extremely small

R-Square means that seasonality only affects a very small amount of spending with males.

So why might this happen? Why might seasonality not affect spending? Carrie et. al highlights in an article that the rise of technology, specifically in e-commerce has changed the way store-based retail functions. Furthermore, these changes in e-commerce have changed traditional shopping habits and trends. These changes could affect the effect of seasonality and spending. The convenience of shopping from home has led many people to spend similar amounts of money throughout the year. For example, both men and women can shop from the convenience of their home in winter and summer.

After learning that seasonality does not affect spending, we decided to take a look at things through a more narrow lens. The types of spending were categorized into luxury (wants) and necessities (needs). The categories under luxury were: eating out, entertainment, and travel whereas the categories under necessities were home items, groceries, and fuel. Our goal was to check whether it was males who spent more than females on luxury items. We believed that women would spend more on luxury items than men because of societal pressure as women may need to spend more to maintain a certain status compared to men. To do so, we conducted hypothesis testing and started by defining our hypotheses :

Ho: There is no difference in the average amount spent on luxury items between males and females (μMale=μFemale).

Alternative hypothesis(Ha) : There is a difference in the average amount spent on luxury between males and females (μMale$\neq\mu$Female).

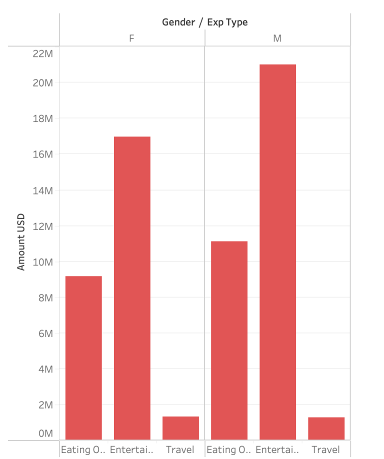
To test out our hypothesis, we conducted a T-test to compare the average spending behaviors of both genders and set α=0.05. The results showed that the T-value is 5.69, P-value is

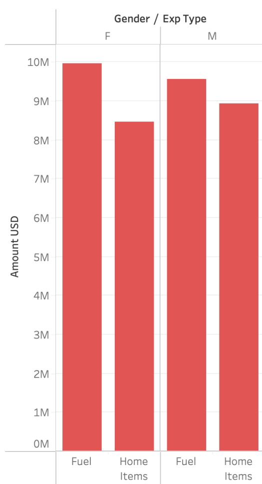
0.0000000128. Based on these findings we are able to reject the null hypothesis, our data shows that there is a difference in the average amount spent in the luxury category with men spending more than females. We then repeated the process to see if there was a difference in the average spending for males and females on necessity items. Our hypothesis were defined in similar fashion :

Ho: There is no difference in the average amount spent on necessity items between males and females (μMale=μFemale).

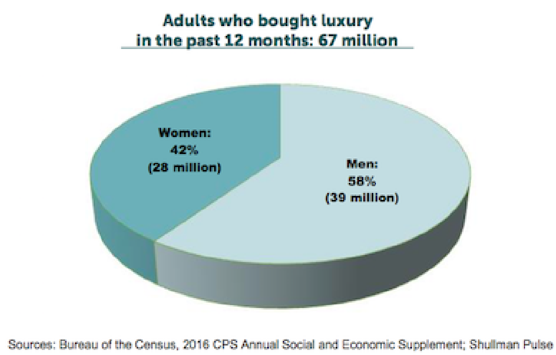
Alternative hypothesis(Ha) is : There is a difference in the average amount spent on necessity between males and females (μMale≠μFemale).

To test out our hypothesis, we repeated the process of conducting a T-test with α=0.05 to measure the behaviors on necessities items between both genders. The test presented us with the results of: T-Value: -1.02, P-Value of 0.31. As the P-Value 0.31 > 0.05. We do not reject the null hypothesis, the data shows that there is no statistically significant difference in necessity spending between males and females. The chart on the right visualizes gender vs luxury spending and the chart on the left visualizes gender vs necessity spending. Males spend more on luxury spending compared to females.





From our study, it can be concluded that males spend more on luxury items than women. To explain our research, an article by Parasi highlighted the percentage of adults who purchased luxury items in 2016 (one of the years our data contained information from) that also showcased



that men spend more than women. A reason for this may be because men are spending on these luxury items for their significant others, they may be paying for dates, buying gifts, and paying for travel. This conclusion helps answer the "Why" from our data. However, it is important to note that we did face limitations during our study. The data did now show if the individual was spending for themselves or others. Another limitation presented was relevant factors such as income, personal preferences, and whether the individual is in a relationship or not were unavailable in the data. All these variables would affect spending behavior. The data we worked with only contained survey results from India, to draw a stronger conclusion spending patterns data from more countries would be required. Finally, economic conditions, hiring patterns, supply chain disruptions, and various policy changes will affect spending patterns that are not highlighted in our data.

Our research concluded that seasonality does not impact spending, likely due to the rise of e-commerce and shifting shopping trends. Regardless of the time of year, shopping with your significant other will result in spending. Notably, we found that men spend more on luxury items, often on gifts and dates, suggesting that men should consider saving up in advance before going out with their girlfriends.

Works Cited

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Parisi, Danny. "Men Spend More on Luxury but Shop the Same Amount as Women: Report." *Luxury Daily*, 24 May 2017